JOSÉ ALBERTO LÓPEZ LÓPEZ

Big Data Engineer | Data Scientist | Machine Learning Engineer

B.S. Physics M.S. Artificial Intelligence

j6.josealberto@gmail.com

ABOUT ME

Big Data Engineer with 10+ years of programming experience, primarily in Python, C++, and Scala; 5+ years of industry experience, mainly in big data and data science projects, developing and maintaining about 15 large-scale data pipelines processing 10M+ records daily, and also building AI models for fraud prevention. Complementary hands-on experience in fields such as computer vision and robotics.

RELEVANT EXPERIENCE

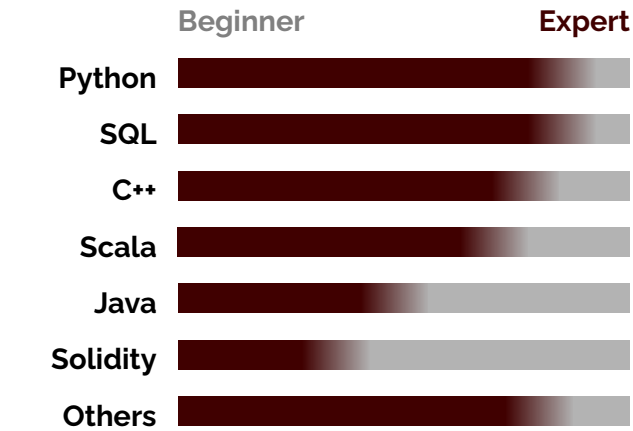
Big Data Engineer
SANTANDER (via INDRA) · Mexico City
dec 2022 – may 2025
Developed and maintained about 15 data pipelines related to bank fraud prevention and ETL processes, using Spark (Scala and Python), and Snowflake (SnowSQL and Snowpark).

Big Data Engineer | AI Developer
GRUPO SALINAS · Mexico City
dec 2021 – aug 2022
Big Data and AI Developer in projects related to bank credit fraud prevention through AI models, using Spark, Hadoop, Kafka, among other technologies.

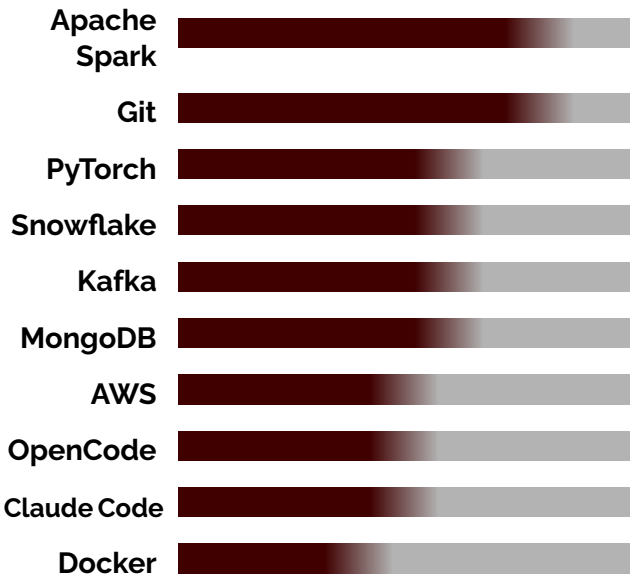
Data Scientist
MIGALA · Mexico City
dec 2021 – feb 2022
Data analysis of a survey conducted among potential members of a new political party. Foundations of Blockchain project with Solidity and other components.

Data Scientist
BEST PRACTICES CONSULTING · Querétaro
mar 2021 – sep 2021
Development of data pipelines and Power BI dashboards in diverse companies projects, including dashboards with gas distribution routes information in real time.

PROGRAMMING LANGUAGES



FRAMEWORKS & TECHNOLOGIES



COMPLEMENTARY EXPERIENCE

- Experience with OpenCode and Claude Code (skills, commands, MCP, permissions, deploying subagents in parallel, etc.).
- Knowledge of LLMs, RAG, and agentic systems internals, how to develop them and techniques to mitigate hallucinations, lack of knowledge and prompt injections.
- Knowledge of MLOps and the machine learning models lifecycle.
- Experience deploying popular AWS services (EC2, S3, SQS, Lambda, CloudFormation, etc.).
- Complementary experience with Keras and TensorFlow (in addition to PyTorch) for deep learning and machine learning.
- Deep knowledge of dense, convolutional, and recurrent neural networks; LLM and transformers architectures; as well as evolutionary and nature-inspired algorithms.
- Robotic Operating System (ROS) with C++ and Python.
- Computer vision with OpenCV library in C++ and Python.
- Experience in parallel programming with OpenMS and MSI libraries in C and C++.

EDUCATION

2020 **Master's in Artificial Intelligence**
ARTIFICIAL INTELLIGENCE RESEARCH CENTER · University of Veracruz, Mexico

2018 **Bachelor's in Physics**
FACULTY OF PHYSICS · University of Veracruz, Mexico
Ranked 1st in class.

LANGUAGES

Spanish Native

English B2

COURSES & CERTIFICATIONS

Provider	Subject	ID and URL
AWS	AWS Cloud Practitioner	
Anthropic Academy	Claude Code	
Udemy	Docker & Kubernetes	
MongoDB Academy	MongoDB	
Udemy	Blockchain	
Udemy	Time Series Data Analysis	
Udemy	Git	

PERSONAL PROJECTS

2026 **docuLATEX** ·
A format and parser for code documentation from TXT to PDF via LaTeX, that produces a nice-looking document.

2019 **Robot Host** ·
Robot host programmed in C++ with ROS. The robot detects diners via face detection and leads them to a suitable table.

2019 **Augmented reality** ·
Augmented reality demo made with Python and OpenCV.

CONTACT

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Alberto López

REFERENCES

Mercedes Potrero Morales
Technology Project Manager at NFQ.
Relation: Previous boss at Indra.
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Antonio Marín Hernández, PhD
Researcher at the Artificial Intelligence Research Center of University of Veracruz, Mexico.
Relation: Master's Thesis director.
anmarin@uv.mx

Marxlenin Zapata Yañez, MS
Chief of Reactor Engineering department at Laguna Verde Nuclear Power Plant.
Relation: Previous boss.
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